Linux Commands – DevOps Practice Guide

Based on the commands shown in your Linux/Vagrant practice history. Use this as a quick revision sheet: understand the purpose, syntax, and a simple example for each command.

ls

List files and directories.

```
Example: ls
```

What happens: Shows the contents of the current directory.

mkdir

Create directories.

```
Example: mkdir testdir
```

What happens: Creates a directory named testdir.

mkdir {1..5}

Brace expansion + mkdir; creates several directories.

```
Example: mkdir testdir{1..5}
```

What happens: Creates testdir1 through testdir5.

rm

Remove files or directories.

```
Example: rm file.txt
```

What happens: Deletes file.txt.

rm -rf

Forcefully remove directories and their contents recursively. Use carefully.

```
Example: rm -rf testdir
```

What happens: Deletes testdir and everything inside it without prompting.

pwd

Print the current working directory.

```
Example: pwd
```

What happens: Shows the absolute path of your current location.

cd

Change directory.

```
Example: cd /tmp
```

What happens: Moves to /tmp.

history

Show previously executed commands.

```
Example: history
```

What happens: Displays the shell command history.

clear

Clear the terminal screen.

```
Example: clear
```

What happens: Removes visible terminal output; it does not delete history.

whoami

Show the current logged-in username.

```
Example: whoami
```

What happens: For example, root or vagrant.

cat

Display file contents; can also concatenate files.

```
Example: cat firstfile.txt
```

What happens: Prints the contents of firstfile.txt.

yum

Package manager commonly used by CentOS/RHEL systems.

```
Example: sudo yum install vim -y
```

What happens: Installs Vim. -y automatically answers yes to prompts.

vim

Terminal text editor.

```
Example: vim firstfile.txt
```

What happens: Opens or creates firstfile.txt in Vim.

sudo

Run a command with elevated privileges.

```
Example: sudo -i
```

What happens: Starts a root login shell if permitted.

hostname

Display the system's hostname.

```
Example: hostname
```

What happens: Shows the machine's host name.

ls -ltr

List in long format, sort by modification time, oldest first.

```
Example: ls -ltr /etc/
```

What happens: Detailed listing of /etc, with older entries first.

cp

Copy files or directories.

```
Example: cp a.txt b.txt
```

What happens: Copies a.txt to b.txt.

Important Options & Syntax

Command	Option	Meaning
ls	-l	Long/detailed listing: permissions, owner, size, date.
ls	-a	Show hidden files/directories.
ls	-h	Human-readable sizes (usually combined with -l).
ls	-t	Sort by modification time.
ls	-r	Reverse the sort order.
cp	-r / -R	Copy directories recursively.
cp	-i	Ask before overwriting.
cp	-v	Show what is being copied.
rm	-r	Remove directories recursively.
rm	-f	Force removal without prompting.
mkdir	-p	Create parent directories when needed.
sudo	-i	Start a root login shell.
yum	-y	Automatically answer yes to prompts.

How to Get Help for Any Linux Command

Method	Purpose	Example
command --help	Quick usage and available options.	cp --help
man command	Detailed manual page.	man cp
info command	Detailed GNU documentation for many commands.	info cp
which command	Show the executable path when applicable.	which cp
type command	Tell whether a command is an alias, builtin, function, or external command.	type cp

Practice Sequence

1. Navigation: pwd → ls → cd
2. Files/directories: mkdir → touch → cat → cp → mv → rm
3. Information: whoami → hostname → history
4. Permissions/admin: sudo → chmod → chown
5. Packages: yum/dnf or apt, depending on the Linux distribution
6. Editors: vim
7. System administration: ps → top → df → du → free

Safety note: Be especially careful with commands such as **rm -rf** and commands executed with **sudo**. Always verify your current directory with **pwd** before deleting or modifying files.